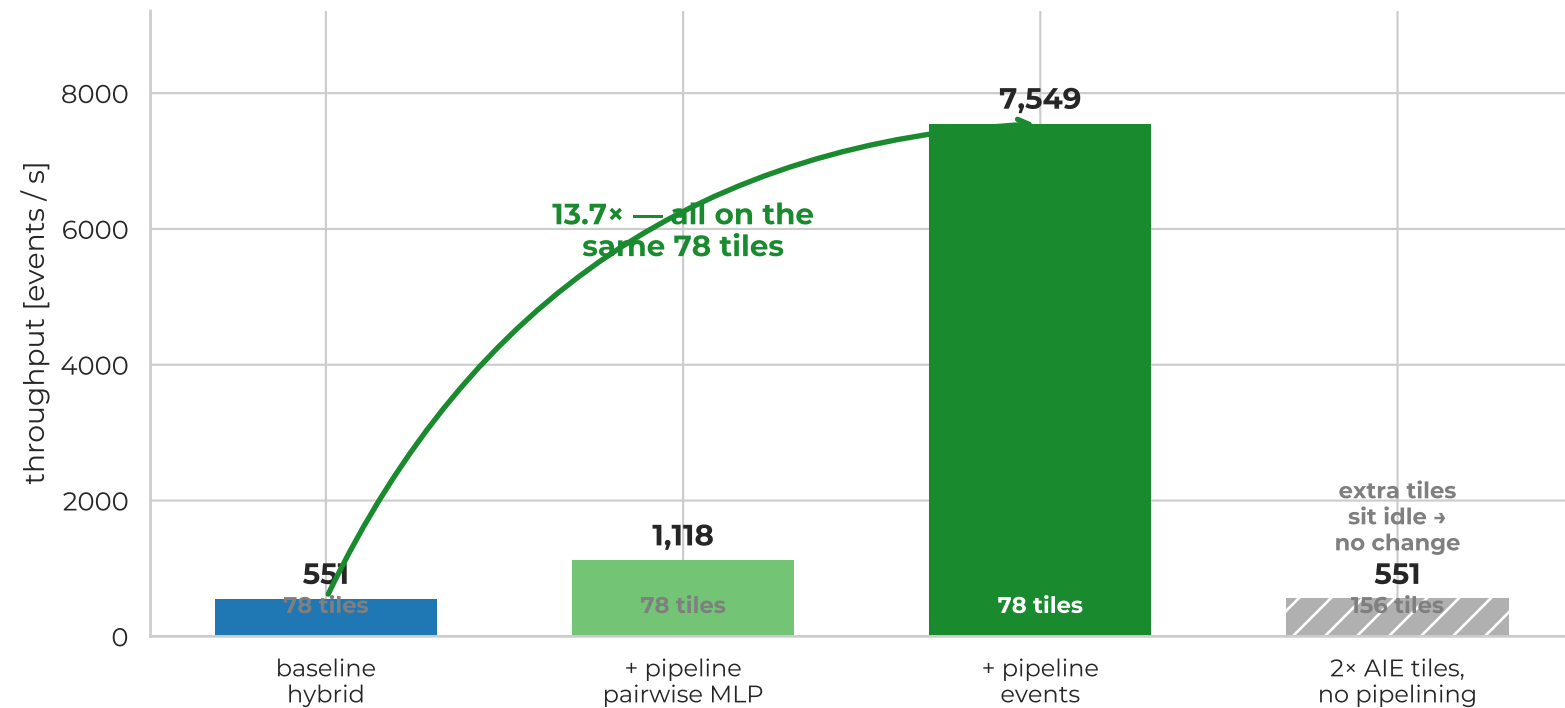
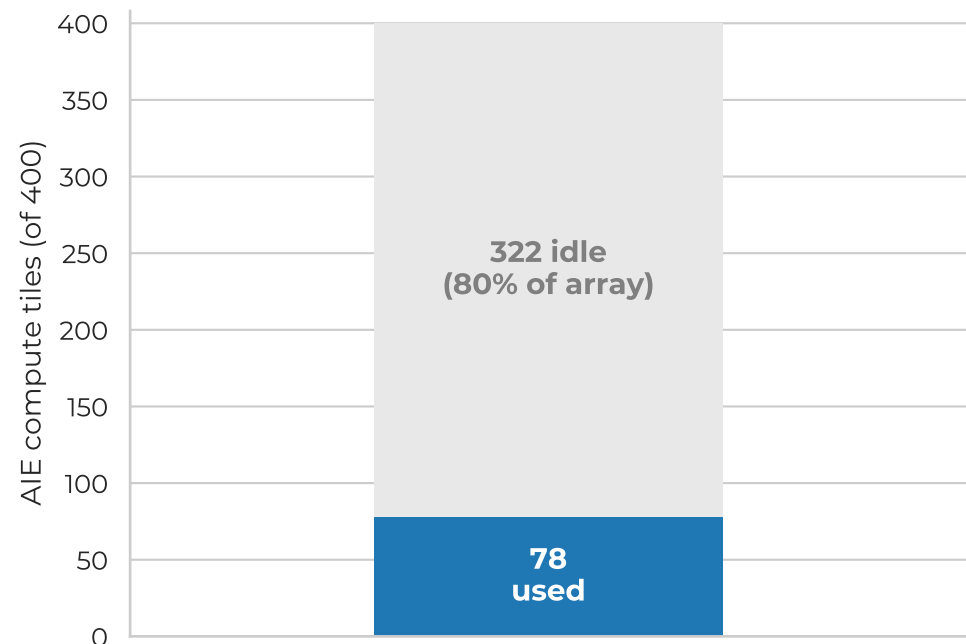


AIE is tile-rich but pipeline-bound — more tiles wouldn't help

Throughput scaled by PIPELINING, not by adding tiles



The array is 81% idle



standalone AIE attention:
obj 4,738 ev/s · cand 14,808 ev/s
→ faster than the pipeline can feed it

Latency of one event is a serial chain (embed→pairwise→obj→...→AE), so more tiles can't shorten it. Throughput is set by the slowest pipeline STAGE (data movement / PL), not by AIE compute—which is why overlapping events on the SAME 78 tiles gave 13.7×, while extra tiles would just idle (and a 2nd full pipeline doesn't fit the PL fabric).